VENKATA RAMIREDDY SEELAM

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Profile Summary

Full-Stack Software Engineer with 2+ years of experience building scalable backend, distributed, and web applications using Python, Java, JavaScript, TypeScript, React, FastAPI, Spring Boot, and AWS. Built production services handling 10K+ daily requests at 99.9% uptime, with experience across REST APIs, databases, distributed systems, cloud infrastructure, CI/CD, testing, and AI-powered applications.

Education

San José State University — San Jose, CA

M.S. in Software Engineering – Applied Data Intelligence | GPA: 3.89/4.0 — *Jan 2025 – Dec 2026*

Experience

Software Engineer – Backend & Infrastructure | *upGrad* — Aug 2023 – Nov 2024

- Built Python/FastAPI services and asynchronous job-processing workflows (submitted → queued → running → completed/failed) for a production video-transcript search system, improving learner engagement by ∼30%.
- Developed REST APIs and JavaScript-based interfaces connecting application workflows with backend services, supporting video processing, search, and natural-language Q&A functionality.
- Deployed and operated containerized services on Kubernetes and AWS handling 10K+ daily requests at ∼99.9% uptime; built centralized observability using Elasticsearch and Kibana, reducing incident-detection time by ∼50%.
- Automated releases via CI/CD using Jenkins and GitHub Actions (test → build → deploy → validate), wrote unit and integration tests, and owned production debugging, incident response, and bug fixes.

Software Engineering Intern | *Darwinbox* — Jan 2023 – Jul 2023

- Developed and optimized backend services using Java and Spring Boot, tuning SQL queries and schema design to improve REST API performance across a high-volume enterprise HR platform.
- Wrote unit and integration tests, debugged application issues, and participated in code reviews as part of an Agile software development lifecycle.

Projects

StreamTide – Distributed Data Processing System | *Python, PySpark, Kafka, AWS EMR, SQL*

- Architected a fault-tolerant distributed system processing 660M+ records using PySpark, Kafka, and AWS EMR with partitioning, parallel processing, and failure handling.
- Implemented Bloom Filters, DGIM, and LSH algorithms from scratch, reducing query latency by 50%.

Grocify – Full-Stack Grocery Management Platform | *React, TypeScript, Python, FastAPI, PostgreSQL, Redis*

- Built a responsive React frontend integrated with FastAPI REST APIs and PostgreSQL for pantry inventory, search, and personalized recommendations and Developed a recommendation engine generating 100+ personalized suggestions across 500+ pantry combinations with sub-200ms response times.

Agentic RAG for Healthcare Accreditation | *Python, LangChain, FastAPI, BGE-M3, Docker, AWS*

- Built an agentic RAG system over a 40K+ page corpus using query routing, hybrid retrieval, BGE-M3 embeddings, and two-stage reranking; deployed the FastAPI service on AWS using Docker and CI/CD.

Leadership Experience & Hackathons

Cloudathon – San José State University — 2026

AWS-powered Cloud Infrastructure Hackathon — *San Jose, CA*

- Led a team of 4 to re-encrypt hundreds of Amazon S3 objects using AWS KMS and migrate an Aurora database to Graviton2 with zero-downtime failover.

Technical Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL
Computer Science: Data Structures, Algorithms, Object-Oriented Programming, Design Patterns, Multithreading/Concurrency
Backend & APIs: REST APIs, FastAPI, Spring Boot, Node.js, Microservices, System Design, API Design
Frontend: React, TypeScript, JavaScript, HTML/CSS, Responsive Web Development
Distributed Systems: Apache Spark, Kafka, Message Queues, Distributed Processing
Cloud & Databases: AWS (EC2, S3, EMR, KMS, Aurora), PostgreSQL, MongoDB, Redis, MySQL
Testing & Infrastructure: Unit/Integration Testing, Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, Git
AI/ML: LLMs, RAG, LangChain, PyTorch, Hugging Face, Model Evaluation